

Digital Manufacturing Value Case & KPI Impact Model

Inputs & Assumptions

Directional model uses fictional baselines and target-state estimates. Values must be validated with site-specific data.

KPI	Baseline	Target	Change	Impact %	Qualitative impact	Confidence
Deviation investigation effort (hrs/month)	120	84	-36	30%	Reduced investigation load	Medium
Batch review cycle time (days)	8	5	-3	38%	Faster review readiness	Medium
Manual documentation effort (hrs/month)	180	105	-75	42%	Less reconciliation	Medium
Right-first-time execution (%)	88	94	+6 pts	7%	Improved execution quality	Medium
Release readiness (%)	72	88	+16 pts	22%	Earlier visibility	Medium
Manufacturing visibility score	55	78	+23	42%	Better decision support	Low-Med
Schedule adherence (%)	82	90	+8 pts	10%	Improved planning signal	Low-Med

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Current State Baseline

Deviation investigation effort, batch review cycle time, manual documentation effort, right-first-time execution, release readiness, manufacturing visibility, and schedule adherence are modeled as baseline indicators.

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Target State Scenario

Target values reflect improved MES execution, EBR workflow clarity, review readiness, integration visibility, and governance.

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KPI Impact Model

Improvement amount = target value - baseline value. Improvement percent = improvement amount / baseline value, direction-adjusted for lower-is-better metrics.

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Value Summary / Dashboard

Highest directional impact: review cycle time, manual documentation effort, release readiness, and visibility.
Confidence is illustrative and should be refined with stakeholder validation.

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Notes & Disclaimer

All numbers are illustrative, fictional, directional, and non-proprietary. This is a portfolio artifact, not a validated financial model.

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